

APPENDIX — FULLY SOLVED REAL PAPER: 11 FEBRUARY 2026, SHIFT 1 (100 Qs)

The real exam as reconstructed from candidate memory (careerpower/Adda247 'Exam Day Based' paper, 11 Feb 2026; official papers are never published). Every question carries: Answer → Trick (the fastest exam-hall method) → Solution (≤ 4 lines) → Trap (the planted distractor). A few stems/options were lost in PDF extraction and are marked; several were recovered from the embedded solutions and re-verified. Question types are interleaved, exactly like the live paper (single section, mixed order — verified against the official response-sheet export for this cycle). The three RC passages sit at Q20–24, Q40–44 and Q90–94.**

Q.1 If the 6-digit number N46M37 is divisible by 11, then which of the options below can give a possible correct relation between M and N?

- A. $M = N$
- B. $M + N = 2$
- C. $M - N = 1$
- D. $M - N = -2$

Answer: D — $N - M = 2$, i.e. $M - N = -2$

Trick: Alternating-sum rule: (sum of digits in odd places) – (sum of digits in even places) must be 0 or ± 11 . Odd places hold N, 6, 3; even places hold 4, M, 7 — so $N + 9 = M + 11$ instantly.

Solution: Odd-place sum = $N + 6 + 3 = N + 9$. Even-place sum = $4 + M + 7 = M + 11$. Set equal: $N + 9 = M + 11 \rightarrow M - N = -2$. (Check: $N = 5$, $M = 3$ gives 546337, divisible by 11.)

Trap: Counting place positions from the left instead of the right flips the odd/even sets and gives $M - N = 2$ — the mirror distractor (option C territory).

Q.2 Two letters are selected from the word 'TENDULKAR'. The probability that both are vowels is:

- A. $2/9$
- B. $1/12$
- C. $5/9$
- D. $5/12$

Answer: B — 3 vowels among 9 letters

Trick: Chance of first pick being a vowel = $3/9 = 1/3$; with one vowel gone, second pick = $2/8 = 1/4$. Multiply fractions — no combinations needed.

Solution: $P = (3/9) \times (2/8) = (1/3) \times (1/4) = 1/12$.

Trap: Using 9×8 as the total (ordered pairs) while keeping favourable outcomes unordered — that yields $3/72 = 1/24$, not offered; the planted wrong pick is $2/9$ (favourable/first-pick only).

Q.3 4 years ago, the age of a father was 37 years more than twice his son's age. After how many years, from now, will he be twice his son's age?

- A. 31
- B. 33
- C. 30
- D. 35

Answer: B — 33 years from now

Trick: The age-gap engine: "father will be twice son's age" happens when $F - 2S = 0$. Whatever constant $F - 2S$ equals today stays fixed forever, so just convert the past condition to $F - 2S = \text{constant}$.

Solution: 4 years ago: $F - 4 = 2(S - 4) + 37 \rightarrow F - 2S = -8 + 37 + 4 = 33$. Father is twice son's age exactly when $F - 2S = 0$, i.e. 33 years from now.

Trap: Forgetting to shift BOTH ages back 4 years (writing $F - 4 = 2S + 37$) gives $F - 2S = 41 \rightarrow$ answer 41, the classic over-shoot distractor.

Q.4 Simplify: $3((5/3)x^2 - 28x + 15) - 5(x^2 + 6x - 15)$

- A. $114x + 120$
- B. $-114x - 120$
- C. $114x - 120$
- D. $-114x + 120$

Answer: D — $-114x + 120$

Trick: Kill x^2 first: $3 \times (5/3)x^2 = 5x^2$ and $-5 \times x^2 = -5x^2$, so the quadratic vanishes. Then only the x-terms and constants remain — combine in one line.

Solution: x-terms: $3(-28x) = -84x$ and $-5(6x) = -30x \rightarrow -114x$. Constants: $3(15) = 45$ and $-5(-15) = +75 \rightarrow 120$. Result: $-114x + 120$.

Trap: Dropping the double negative on $-5(-15) = +75$ gives $-114x - 120$ — option B's planted error.

Q.5 In $\triangle ABC$, $DE \parallel AC$, where D and E are the points on sides AB and BC, respectively. If $BD = 2$ cm and $AD = 19$ cm, then what is the ratio of the area of $\triangle BDE$ to the trapezium ADEC?

A. 12 : 435

B. 10 : 11

C. 4 : 437

D. 13 : 8

Answer: C — small triangle : big trapezium

Trick: Same-angle triangles $\rightarrow$ area scales as the SQUARE of the base ratio. $BD : BA = 2 : 21$, so areas are 4 : 441; subtract to get the strip without any formula recall.

Solution: $BA = 2 + 19 = 21$. $\triangle BDE : \triangle BAC = (2/21)^2 = 4 : 441$. Trapezium ADEC = $441 - 4 = 437$. So $\triangle BDE : ADEC = 4 : 437$.

Trap: Comparing with $BD : AD = 2 : 19$ (squaring to 4 : 361) — the trapezium sits on $BA = 21$, not on the cut-off piece $AD = 19$.

Q.6 In the following triads, each group of letters is related to the subsequent one following a certain logic. Select from the given options, the one which follows the same logic.

LIVE – ELVI – LEIV

HANG – GHNA – HGAN

A. PORT – TPOR – ORTP

B. REDS – ERDS – SEDR

C. DUSK – KDSU – DKUS

D. PAIN – NPIA – PANI

Answer: C — DUSK – KDSU – DKUS

Trick: Tag positions of the first word, then map each later word to a position sequence; the option whose three sequences match the model wins — no letter-shuffling by eye.

Solution: LIVE $\rightarrow$ ELVI reads 4-1-3-2; ELVI $\rightarrow$ LEIV reads 2-1-4-3. HANG $\rightarrow$ GHNA = 4-1-3-2 and GHNA $\rightarrow$ HGAN = 2-1-4-3. Only DUSK $\rightarrow$ KDSU (4-1-3-2) $\rightarrow$ DKUS (2-1-4-3) repeats both steps exactly.

Trap: PAIN – NPIA matches the FIRST transformation (4-1-3-2) but fails the second — options that copy only step 1 are the planted decoys.

Q.7 If '+' means 'division', ' $\div$ ' means 'multiplication', '-' means 'addition' and ' $\times$ ' means 'subtraction', then find the value of

$$15 + 3 - 8 \div 9 \times 7$$

A. 81

B. 64

C. 72

D. 70

Answer: D — 70

Trick: Substitute signs directly, then apply BODMAS strictly: $+$ $\rightarrow$ $\div$, $\div$ $\rightarrow$ $\times$, $-$ $\rightarrow$ $+$, $\times$ $\rightarrow$ $-$. So the string becomes $15 \div 3 + 8 \times 9 - 7$.

Solution: $15 \div 3 + 8 \times 9 - 7 = 5 + 72 - 7 = 70$.

Trap: Reading left to right ignoring BODMAS: $(15 \div 3 + 8) \times 9 - 7 = 200$ — nowhere near the options; near-miss 72 comes from computing 8×9 before $15 \div 3$ and mis-tracking the $+7/-7$ flip.

Q.8 X, Y and Z invested a sum in the ratio of 58 : 72 : 20, respectively. If they earned a total profit of Rs. 3,750 at the end of the year, what is the difference between the shares of Y and Z?

A. Rs. 1,300

B. Rs. 1,404

C. Rs. 1,231

D. Rs. 1,254

Answer: A — Rs. 1,300

Trick: Never compute two shares: difference = (ratio gap $\div$ ratio total) $\times$ profit = $(52/150) \times 3750$, one multiplication.

Solution: $3750/150 = 25$ per part. $Y - Z = (72 - 20) \times 25 = 52 \times 25 = 1300$.

Trap: Using Y's share alone ($72 \times 25 = 1800$) or Z's (500) and forgetting to subtract — or dividing by 100 instead of 150, giving 1560.

Q.9 Kings and Queens of black colour are taken out from a deck of 52 playing cards. A card is drawn from the remaining well-shuffled cards. Probability of getting a spade card is:

- A. $1/4$
- B. $11/13$
- C. $11/48$
- D. $11/52$

Answer: C — 11 spades among 48 cards

Trick: Update both numerator and denominator: black K + Q removes 2 spades (of 13) and 4 cards (of 52). Spades are untouched by "black removal" beyond those two faces — the other 11 remain.

Solution: Favourable = $13 - 2 = 11$; total = $52 - 4 = 48$. $P = 11/48$.

Trap: Keeping the deck at 52 ($11/52$, option D) or forgetting that 2 spades left with the removed black kings/queens ($13/48$).

Q.10 3 years ago from now, the age of father was 14 years more than twice his son's age. After how many years, from now, will he be twice his son's age?

- A. 12
- B. 14
- C. 8
- D. 11

Answer: D — 11 years from now

Trick: Same gap engine as Q3: convert the past statement to $F - 2S = \text{constant}$; "twice" arrives exactly when that constant counts down to zero.

Solution: $F - 3 = 2(S - 3) + 14 \rightarrow F - 2S = 14 - 6 + 3 = 11$. Father is twice son's age 11 years from now.

Trap: Writing $F - 3 = 2S + 14$ (forgetting the son's -3) $\rightarrow$ constant 17 $\rightarrow$ answer 17, and misreading "3 years ago" as "after 3 years" gives 5 — both classic slips; option B (14) baits those who reuse the raw 14.

Q.11 The profit made on a product is 66%. What is the loss percentage (rounded off to two decimal places) incurred on the sale if the figures of both the cost price and the selling price are interchanged?

- A. 39.76%
- B. 38.40%
- C. 41.12%
- D. 38.57%

Answer: A — 39.76%

Trick: Base-100: $CP = 100 \rightarrow SP = 166$. Swap and the loss % is just $66/166$ — a single fraction, no algebra. Memory hook: $p/(100 + p)$.

Solution: $66/166 \times 100 = 39.759... \approx 39.76\%$.

Trap: Computing $66/100 = 66\%$ or $66/166$ rounded carelessly to $39.75/38.57$ — the trap is dividing by the NEW SP (100) instead of the new CP (166).

Q.12 The mean proportional between $12 + 3\sqrt{4}$ and $16 - 4\sqrt{4}$ is:

- A. 7
- B. 22
- C. 19
- D. 12

Answer: D — 12

Trick: Mean proportional = $\sqrt{ab}$. Simplify the radicals first: $\sqrt{4} = 2$, so the pair collapses to 18 and 8 — then $18 \times 8 = 144$ is a perfect square by design.

Solution: $12 + 3\sqrt{4} = 18$; $16 - 4\sqrt{4} = 8$. Mean proportional = $\sqrt{(18 \times 8)} = \sqrt{144} = 12$.

Trap: Taking $\sqrt{4}$ as 4 (giving 24 and 0, or $12 + 12 = 24$ and $16 - 16 = 0$ — nonsense values) instead of 2; also, plain arithmetic mean $(18 + 8)/2 = 13$ is the near-miss.

Q.13 A man spends 28% of his monthly salary on house rent. If every month, he also spends ₹687 on conveyance and ₹7,483 on grocery and saves the remaining ₹218, his monthly salary is:

- A. ₹11,738

- B. ₹11,650
C. ₹11,594
D. ₹11,743

Answer: B — ₹11,650

Trick: Bundle the fixed spends: everything except rent = $687 + 7483 + 218 = 8388$, and that must be the 72% left after rent. So salary = $8388 \times (100/72)$.

Solution: $100/72 = 25/18$; $8388/18 = 466$; $466 \times 25 = 11,650$.

Trap: Taking 8388 as 28% (rent instead of the remainder): $8388/0.28 = 29,957$ — absurd; or adding 218 to expenses instead of treating it as savings.

Q.14 Point A is 18 m to the east of Point B. Point B is 22 m to the north of Point C. Point D is 11 m to the west of Point C. Point E is 11 m to the south of point D. Point X is 29 m to the east of Point E.

What is the shortest distance between Point A and Point X?

- A. 11 m
B. 33 m
C. 21 m
D. 10 m

Answer: B — 33 m

Trick: Assign coordinates from a fixed origin and cancel equal-and-opposite legs before any Pythagoras — most direction questions collapse to one straight line.

Solution: Take C at (0, 0): B = (0, 22), A = (18, 22); D = (-11, 0), E = (-11, -11), X = (18, -11). A and X share $x = 18$, so $AX = 22 - (-11) = 33$ m.

Trap: Drawing X 29 m east of D instead of E, or adding the north-south legs with a sign error — then Pythagoras gives a wrong $\sqrt{\text{—}}$ -answer instead of the clean 33.

Q.15 The diameter of the base and slant height of a right circular cone are 30 cm and 113 cm, respectively. Find the volume (in cm^3) of the given cone. (Use $\pi = 22/7$)

- A. 26,360
B. 26,375
C. 26,400
D. 26,435

Answer: C — 26,400 cm^3

Trick: Pull h from $l^2 - r^2 = 113^2 - 15^2$; both are from Pythagorean-style partners — $113^2 - 15^2 = (113 - 15)(113 + 15) = 98 \times 128 = 12544 = 112^2$. No long squaring needed.

Solution: $r = 15$, $h = \sqrt{98 \times 128} = 112$. $V = (1/3)(22/7)(15^2)(112) = (22/7)(225)(112/3) = (22/7)(75)(112) = 22 \times 75 \times 16 = 26,400 \text{ cm}^3$.

Trap: Using 30 as the radius in $V = (1/3)\pi r^2 h$ (four-fold error $\rightarrow 1,05,600$) or computing h from $113^2 - 30^2$ instead of 15^2 .

Q.16 A chord AB is drawn in a circle with centre O and radius 6 cm. If the shortest distance between the centre and the chord is 4.5 cm, then find the length of chord AB.

- A. $2\sqrt{7}$ cm
B. $4\sqrt{7}$ cm
C. $3\sqrt{7}$ cm
D. $\sqrt{7}$ cm

Answer: C — $3\sqrt{7}$ cm

Trick: Half-chord = $\sqrt{r^2 - d^2}$. Work with quarters when d is a half-integer: $r^2 - d^2 = 36 - 20.25 = 15.75 = 63/4$, so half-chord = $\sqrt{63/2} = 3\sqrt{7}/2$.

Solution: $(AB/2)^2 = 6^2 - 4.5^2 = 36 - 20.25 = 15.75$. $AB = 2\sqrt{63/4} = \sqrt{63} = 3\sqrt{7}$ cm.

Trap: Reporting the half-chord ($3\sqrt{7}/2 \approx$ option A $2\sqrt{7}$ territory) — forgetting to double; or computing $36 - 4.5$ (forgetting to square d) $\rightarrow AB = 2\sqrt{31.5}$.

Q.17 Each of P, Q, R, S, T, U and V has an exam on a different day of a week starting from Monday and ending on Sunday of the same week. P has the exam on Tuesday. Only two people have exams after R. S has the exam immediately before Q but after R. U has the exam immediately after V. How many people have exams between T and S?

- A. One

- B. Three
- C. Four
- D. Two

Answer: C — Four

Trick: Fix the anchored days first (P = Tue; R = Fri because only 2 follow), then place the VU pair in the only two adjacent free slots — Mon–Wed — and S–Q (Sat–Sun) is forced by "S after R".

Solution: P = Tue, R = Fri, S = Sat, Q = Sun. VU pair must be adjacent: V = Wed, U = Thu leaves T = Mon (unique). Between T (Mon) and S (Sat): Tue, Wed, Thu, Fri = 4 people.

Trap: Placing the V–U pair on Mon–Tue (occupying P's Tuesday) — re-check P = Tue; and counting "between" inclusively (5) instead of strictly between (4).

Q.18 Which of the given numbers will replace the question mark (?) in the following series to make it logically complete?

235, 246, 268, 301, 345, ?

- A. 401
- B. 400
- C. 410
- D. 445

Answer: B — 400

Trick: Take differences first: 11, 22, 33, 44 — multiples of 11 climbing by one step. Next difference 55 lands on 400.

Solution: $235 + 11 = 246$, $+22 = 268$, $+33 = 301$, $+44 = 345$, $+55 = 400$.

Trap: Seeing only "differences grow by 11" but starting the ladder wrongly (10, 22, 33 ...) or adding 66 (the next multiple after 44 used twice) → 411/445-type distractors.

Q.19 If letters of the English alphabet are written in reverse order then which letter is 5th to the right of 12th letter from the right?

- A. L
- B. G
- C. O
- D. J

Answer: B — G

Trick: Both references point from the right end, so add: $(12 + 5) = 17$ th from the right of the reversed alphabet. In reverse order Z is 1st, so position $17 = 26 - 17 + 1 = 10$ th forward = J... but wait — reverse-order position p from the right is the letter $27 - p$ forward only when counting from the LEFT; from the RIGHT of a reversed row it is simply the 17th letter of Z Y X W ...

Solution: Reversed: Z Y X W V U T S R Q P O N M L K J I H G F E D C B A. 12th from right = L; 5 more to the right (towards A): K, J, I, H, G → G.

Trap: Adding $12 + 5 = 17$ and converting to the forward 10th letter (J) — mixing left/right reference frames; "5th to the right" here means 5 further from the right end.

Q.20 What is the main idea of the passage?

Read the passage and answer the following questions.

When a machine "learns," it does not learn like a student memorising notes. It learns by adjusting patterns. Consider a recommendation system that suggests videos or articles. At first, it has only rough signals: what you clicked, how long you stayed, what you skipped. From these traces it builds a probability model—an internal guess about what might

keep you engaged. The more it predicts correctly, the more confident it becomes, and the more it shapes what you see

next. This design creates a quiet feedback loop. If the system shows you content similar to what you already watched,

you are more likely to watch it again, which then "confirms" the system's guess. Gradually, the range of options can narrow—not because alternatives disappear, but because they are rarely offered. In such a loop, the system is not neutral. It actively organises attention. The same mechanism that helps you discover relevant material can also trap you in repetitive preferences or amplify extreme content that triggers strong reactions. Developers try to break this loop with "diversity" measures, inserting unfamiliar items to test whether your interests are broader than your history suggests. But this introduces a tension: people often say they want variety, yet they frequently reward familiarity with longer watch time. Platforms therefore face a problem-solution choice. If they optimise only for

engagement, they risk creating echo chambers. If they optimise for exploration and well-being, they may lose short-term clicks. The central question becomes ethical as well as technical: should a system maximise what users consume, or help users choose more consciously?

- A. Recommendation systems are harmless because they only show random content
- B. The passage explains how recommendation systems create feedback loops that can narrow choices, and why balancing engagement with diversity is an ethical problem
- C. The passage argues that users have no role in shaping what they see online
- D. The passage claims that "learning" in machines is identical to human learning

Answer: B — feedback loops + the engagement-vs-diversity ethical tension

Trick: For "main idea", the correct option must cover BOTH halves of the passage: mechanism (feedback loop narrowing choices) AND the closing dilemma (engagement vs diversity/ethics). Options covering only one half, or contradicting a stated line, are out.

Solution: B alone captures the loop (paras 1–2) and the ethical problem-solution framing (para 3). A, C, D each contradict explicit lines (random content; user clicks central; "does not learn like a student").

Trap: Picking the extreme-word options ("harmless", "no role", "identical") — RC main-idea options with absolutes are almost always distractors.

Q.21 Which detail best supports the idea of a "feedback loop"?

Read the passage and answer the following questions.

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- A. The system uses electricity to run servers
- B. Watching similar content makes you more likely to watch it again, which then confirms the system's guess and shapes future suggestions
- C. Users always prefer unfamiliar content
- D. Platforms have no access to user behaviour

Answer: B — the recommendation → repeat viewing → confirmation cycle

Trick: "Supports the idea of X" questions want the sentence that contains the CAUSAL CYCLE, not a fact merely adjacent to the topic. Look for the circular chain in the passage's own words.

Solution: Para 2 states the loop verbatim: similar content → watch again → "confirms" the guess → shapes what you see next. That is option B.

Trap: Choosing A (on-topic but irrelevant mechanism) — topical proximity without causal loop is the planted decoy.

Q.22 Which situation best reflects the passage's warning about "amplifying extreme content"?

Read the passage and answer the following questions.

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A. A library displays books in alphabetical order

B. A platform promotes sensational videos because they generate strong reactions, leading the system to recommend more of the same

C. A teacher gives students different topics to expand knowledge

D. A website removes recommendation features entirely to reduce data use

Answer: B — reaction-driven promotion → more of the same

Trick: Application questions mirror the passage's MECHANISM in a new scenario: find the option where a trigger (strong reactions) is rewarded by the system, creating more of the same.

Solution: The passage: “amplify extreme content that triggers strong reactions.” Option B: sensational videos get promoted for exactly that reason. Same mechanism, new setting.

Trap: C looks “positive” but describes diversity — the OPPOSITE of amplification; positive-tone options are frequent traps in application RC.

Q.23 What is the organisational structure of the passage?

Read the passage and answer the following questions.

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A. Problem and solution: it identifies echo-chamber risk and proposes diversity measures as a partial remedy, then discusses trade-offs

B. Chronological history of the internet from 1990 to 2020

C. Compare and contrast of two different languages

D. A mythological story explaining human curiosity

Answer: A — problem → attempted solution → trade-off analysis

Trick: Map the paragraph openings: para 1 = mechanism, para 2 = the problem it creates, para 3 = “Developers try to break this loop...” — the “try to + tension” signal marks problem–solution structure.

Solution: Para 2 states the problem (narrowing, echo chambers); para 3 offers diversity measures and immediately weighs the trade-off (engagement vs exploration). That is option A.

Trap: Ignoring the final trade-off sentence and calling it pure problem–solution without analysis — but among the options only A fits; B/C/D are absurd distractors.

Q.24 The style of the passage is best described as:

Read the passage and answer the following questions.

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A. Technical yet reflective, using examples to explain an ethical dilemma

B. Sarcastic and ridiculing technology companies

C. Purely narrative with characters and dialogue

D. Emotional and celebratory

Answer: A — technical-reflective with a running example

Trick: Style questions are eliminated first: drop any option needing elements the passage lacks (dialogue, mockery, celebration). Among survivors, match the register — explanatory + evaluative.

Solution: The passage explains a mechanism (signals, probability model) via one sustained example and ends reflecting on an ethical choice — technical yet reflective = A.

Trap: Marking “technical” as a negative and rejecting A — the passage IS technical in vocabulary but plain in tone; B/C/D fail on absence of sarcasm, characters, and emotion respectively.

Q.25 What should come in place of the question mark ? in the given series based on the English alphabetical order?
ZFN, FLT, LRZ, RXF, ?

A. XSD

B. XLD

C. XDL

D. XDS

Answer: C — XDL

Trick: Treat each column of the letter-triplets as its own independent series with a constant jump (+6 here) — three tiny series instead of one big puzzle.

Solution: 1st letters: Z→F→L→R→X (+6 each). 2nd: F→L→R→X→D (+6). 3rd: N→T→Z→F→L (+6, wrapping). Next term: XDL.

Trap: Applying +6 to only the first letter and guessing the rest (XSD/XDS), or forgetting the Z→F wrap-around (giving XF...-type slips).

Q.26 Based on the English alphabetical order, three of the following four letter-clusters are alike in a certain way and thus form a group. Which letter-cluster DOES NOT belong to that group?

(Note: The odd one out is not based on the number of consonants/vowels or their position in the letter-cluster.)

A. BVQN

B. EYTQ

C. AUPM

D. FZVR

Answer: D — FZVR

Trick: For odd-one-out clusters, compute consecutive letter-gaps (with wrap-around) for all four in one pass; three identical gap-signatures expose the odd one instantly.

Solution: BVQN, EYTQ, AUPM all step -6, -5, -3 (cyclically). FZVR steps -6, -4, -4 — its 2nd and 3rd gaps break the pattern.

Trap: Checking only the first gap (all four show -6) and concluding "all alike" — compare every gap position; also avoid vowel/consonant counting, explicitly ruled out by the note.

Q.27 The following pie-chart shows the percentage distribution of the expenditure incurred in manufacturing a ceiling fan. Study the pie-chart and answer the question.

(Transportation = 9%; other heads: Manufacturing 30%, Advertisement 11%, Direct Material 22%, Direct Labour 28%.)

If 10,000 ceiling fans are manufactured and the transportation cost on them amounts to ₹36,000, then what should the selling price of each ceiling fan be, so that the manufacturer can earn a profit of 50%?

- A. ₹40
- B. ₹75
- C. ₹50
- D. ₹60

Answer: D — ₹60

Trick: One anchor percent does everything: $9\% = 36,000 \rightarrow 1\% = 4,000 \rightarrow \text{total cost} = 4,00,000 \rightarrow \text{per fan} = 40 \rightarrow \text{SP} = 40 \times 3/2$. The 150% multiplier ($\times 3/2$) is instant for 50% profit.

Solution: Total cost = $36,000 \times (100/9) = 4,00,000$; per fan = ₹40. SP at 50% profit = $40 \times 1.5 = ₹60$.

Trap: Applying 50% profit to the TOTAL ($6,00,000/10,000 = 60$) coincidentally matches, but the real trap is treating 36,000 as the per-fan base or adding profit as 50% of the selling price ($\text{SP} = 80$).

Q.28 In a code language, TEMPLE is written as 053625. What is the code for PRAY?

- A. 6815
- B. 6814
- C. 6816
- D. 6915

Answer: A — 6815

Trick: Match code length to word length: TEMPLE (6 letters) $\rightarrow$ 6 digits but PRAY (4 letters) needs 4 — so the code is the UNIT digit of each letter's alphabet position (T = 20 $\rightarrow$ 0 is the giveaway).

Solution: P = 16 $\rightarrow$ 6, R = 18 $\rightarrow$ 8, A = 1 $\rightarrow$ 1, Y = 25 $\rightarrow$ 5. PRAY = 6815.

Trap: Using full position values (16-18-1-25) or assuming every letter maps to a 2-digit code — the T $\rightarrow$ 0 digit proves the unit-digit rule.

Q.29 The distance between two stations A and B is 420 km. A train starts from A at 3 p.m. and travels towards B at a speed of 60 km/h. Another train starts from B at 4 p.m. and travels towards A at a speed of 30 km/h. At what time do they meet?

- A. 10 p.m.
- B. 10:30 p.m.
- C. 9 p.m.
- D. 8 p.m.

Answer: D — 8 p.m.

Trick: Head-start first, then relative speed: the early train banks 60 km solo, so only 360 km remain to be closed at 90 km/h — a clean 4 h after 4 p.m.

Solution: By 4 p.m. train 1 has covered 60 km; gap = 360 km. Combined speed = 90 km/h $\rightarrow$ 4 h $\rightarrow$ meet at 8 p.m.

Trap: Skipping the head-start: $420/90 = 4.67$ h after 3 p.m. $\approx$ 7:40 p.m. (or rounding to 10 p.m. style errors); also adding the 1 h AFTER computing the meeting time (9 p.m.) is the classic double-count.

Q.30 Below are given statements followed by some conclusions. You have to take the given statements to be true even if they seem to be at variance with the commonly known facts and then decide which of the given conclusions logically follow(s) from the given statements.

Statements: 1. Some Editors are Actors. 2. All the Actors are Writers.

Conclusions: I. Some Editors are Writers. II. No Actor is Editor.

- A. Neither Conclusion I nor II follows.
- B. Only Conclusion II follows.
- C. Only Conclusion I follows.
- D. Either Conclusion I or II follows.

Answer: C — Only Conclusion I follows

Trick: Chain "Some A are B + All B are C" always yields "Some A are C" — guaranteed conversion. A "No" conclusion can never follow from an all-positive pair of statements.

Solution: I: Editors $\cap$ Actors $\neq \emptyset$ and Actors $\subseteq$ Writers, so some editors are writers — follows. II: directly contradicts "Some Editors are Actors" — fails. Only I follows.

Trap: Choosing "Either I or II follows" because the two conclusions look complementary — they are not: I is affirmed true, so the either-or rule (which needs both individually in doubt) never activates.

Q.31 Let AB and CD be two parallel lines and PQ be a transversal such that PQ intersects AB at the point R and CD at the point S, respectively. If $\angle BRP = (2x + 13)^\circ$ and $\angle DSP = (3x - 22)^\circ$, then find $\angle CSP$.

- A. 105°
- B. 95°
- C. 97°
- D. 83°

Answer: C — 97°

Trick: $\angle BRP$ and $\angle DSP$ are alternate interior angles (both open toward P along the transversal) — set them equal, get x, then $\angle CSP$ is simply the supplement of $\angle DSP$ on line CD.

Solution: $2x + 13 = 3x - 22 \rightarrow x = 35$. $\angle DSP = 3(35) - 22 = 83^\circ$. $\angle CSP = 180^\circ - 83^\circ = 97^\circ$.

Trap: Stopping at 83° (option D) — finding $\angle DSP$ and forgetting the question asks for its supplement; or treating the pair as co-interior (sum 180°) $\rightarrow x = 189/5$, non-integer, a signal you mis-identified the pair.

Q.32 What is the mode of the data given below? [Give your answer correct to 2 decimal places.]

Age in years: 10-20, 20-30, 30-40, 40-50, 50-60, 60-70, 70-80

No. of patients: 29, 35, 25, 10, 18, 34, 17

- A. 39.44
- B. 6.56
- C. 23.75
- D. 22.58

Answer: C — 23.75

Trick: Modal class = highest frequency = 35 (20-30). Then mode = $l + [(f_1 - f_0)/(2f_1 - f_0 - f_2)] \times h$ with $l = 20$, $f_1 = 35$, $f_0 = 29$, $f_2 = 25$, $h = 10$ — the fraction is $6/16$, so $10 \times 6/16 = 3.75$.

Solution: Mode = $20 + (35 - 29)/(2 \times 35 - 29 - 25) \times 10 = 20 + (6/16) \times 10 = 20 + 3.75 = 23.75$.

Trap: Taking 34 (in the 60-70 row) as the peak by scanning the wrong column, or computing $(f_1 - f_0)/(f_1 - f_2) = 6/10 \rightarrow 26$ — dropping the $2f_1$ term is the standard formula slip.

Q.33 The average of eight numbers is 14. The average of five of these numbers is 14. The average of the remaining three numbers is:

- A. 13
- B. 16
- C. 14
- D. 15

Answer: C — 14

Trick: If a subset has the SAME average as the whole, the rest inherit it — leftover sum $112 - 70 = 42$ over $3 = 14$ with zero computation beyond one multiplication each.

Solution: Total = $8 \times 14 = 112$; five known = $5 \times 14 = 70$; remainder = $42/3 = 14$.

Trap: Assuming the "remaining three" must differ (guessing 13 or 15) — when averages match, the complement matches too.

Q.34 Find the value of $[(0.2)^3 + (0.04)^3] \div [(0.1)^2 \times 0.1 + (0.02)^2 \times 0.02]$

- A. 9
- B. 6
- C. 7
- D. 8

Answer: D — 8

Trick: Cube-of-cube structure: write everything as powers of 0.2 — $0.04 = 0.2^2$, $0.1 = 0.2/2$. Numerator = $a^3 + a^6 = a^3(1 + a^3)$; denominator = $a^3(1/8 + 1/64) \times \dots$ — simplest is direct decimal cubes: $0.008 + 0.000064$ over $0.001 + 0.000008$.

Solution: Num = $0.008 + 0.000064 = 0.008064$. Den = $0.001 + 0.000008 = 0.001008$. Ratio = $0.008064/0.001008 = 8$.

Trap: Reading $(0.1)^2 \times 0.1$ as $(0.1)^3 = 0.001$ is fine, but squaring instead of cubing 0.04 (0.0016 vs 0.000064) wrecks the ratio toward 6–7; misplaced decimal points give 80/0.8.

Q.35 Seven boxes A, B, C, D, E, F and G are kept one over the other but not necessarily in the same order. No box is kept above F. Only three boxes are kept between F and E. Only one box is kept between G and C. C is kept immediately above E. Only four boxes are kept between G and A. B is kept at some place above D. How many boxes are kept above E?

- A. Two
- B. Four
- C. One
- D. Three

Answer: B — Four

Trick: Fix the extreme first: F at the top (7). "Three between F and E" + "C immediately above E" forces $E = 3$, $C = 4$ — anchors before neighbours.

Solution: $F = 7 \rightarrow E = 3$, $C = 4$. G–C gap of one $\rightarrow G = 6$. G–A gap of four $\rightarrow A = 1$. D and B fill 2, 5 with B above D $\rightarrow B = 5$, $D = 2$. Above E: positions 4–7 = 4 boxes.

Trap: Mirroring the stack ($E = 5$ with three boxes between E and F reading downward) — "kept between" is symmetric but "C immediately above E" and "no box above F" break the mirror; also counting floor 3 itself in "above E" (5 instead of 4).

Q.36 The diameters of the bases of two cones are equal. If their slant heights are in the ratio of 3 : 4, then what will the ratio of their curved surface areas be?

- A. 19:16
- B. 4:3
- C. 6:9
- D. 3:4

Answer: D — 3 : 4

Trick: $CSA = \pi r l$. Equal diameters kill the r , π cancels — the ratio passes straight through from the slant heights. Any "equal base" CSA/LSA question is a pass-through ratio.

Solution: $CSA_1 : CSA_2 = \pi r l_1 : \pi r l_2 = l_1 : l_2 = 3 : 4$.

Trap: Squaring the ratio (9 : 16) — that rule belongs to areas scaling with $length^2$ (similar figures), not to a linear formula where one factor is already equal.

Q.37 The following line graph shows the population (in million) of Istanbul over 6 years (from 2013 to 2018). Answer the given question based on the line graph.

(Values: 2013 = 1.3 million, 2018 = 2.8 million.)

What is the approximate percentage increase in the population of Istanbul from 2013 to 2018?

- A. 130%
- B. 120%
- C. 118%
- D. 115%

Answer: D — 115%

Trick: Percentage change = change $\div$ ORIGINAL (the earlier year). $1.5/1.3 = 15/13 \approx 1.15 \rightarrow 115\%$. Sanity band: the increase (1.5) is a bit more than the base (1.3), so the answer sits just above 100%.

Solution: $(2.8 - 1.3)/1.3 \times 100 = 1.5/1.3 \times 100 \approx 115.38\% \approx 115\%$.

Trap: Dividing by the FINAL value ($1.5/2.8 \approx 54\%$) or by the change itself (100%) — the "approximate" wording hides option C (118%) as a rounding decoy for those who compute 1.5/1.27-type misreads.

Q.38 In a certain code language, STUDEFARM is written as FGHQSNEZ. How will NUCLEAR be written as in that language?

- A. FLQYVNE
- B. GSQUCNE
- C. AHPYRNE
- D. DKQTCNE

Answer: C — AHPYRNE

Trick: Spot one anchor letter first: S (19) $\rightarrow$ F (6) is +13 — the half-alphabet shift. Confirm with M (13) $\rightarrow$ Z (26), then every letter is +13 with wraparound (an Atbash-complement code).

Solution: N→A, U→H, C→P, L→Y, E→R, A→N, R→E. NUCLEAR = AHPYRNE.

Trap: Applying -13 instead of +13 (gives ANPYRNE-ish strings) or shifting only some letters; also note +13 and -13 differ here because 26-letter wrap is not self-inverse for odd offsets — verify with S→F, not assume.

Q.39 A and B walk around a circular path of 10 km in circumference, starting together from the same point in the same direction. If their speeds are 3 km/h and 2 km/h respectively, after how many hours will they be again at the starting point?

- A. 5 h
- B. 2 h
- C. 10 h
- D. 3 h

Answer: C — 10 h

Trick: "Again at the starting point" = LCM of individual lap times, not the meeting-anywhere time. Lap times are 10/3 h and 5 h → LCM of fractions = LCM(numerators)/HCF(denominators) = 10/1.

Solution: $t_A = 10/3$ h, $t_B = 5$ h. $\text{LCM}(10/3, 5/1) = \text{LCM}(10, 5)/\text{HCF}(3, 1) = 10$ h. Check: A does 3 laps, B does 2 laps — both integer, both at start.

Trap: Using the relative-speed meeting time $(10/(3-2)) = 10$ h — same answer here by luck, but the question is about the STARTING POINT, so the LCM route is the reliable one); choosing 5 h (only B's lap time) or 30 h (LCM of 10 and 15 without dividing by HCF 3).

Q.40 What is the main idea of the passage?

Read the passage and answer the following questions.

In the early decades of the 20th century, a new kind of "map" began to reshape how cities were governed. Instead of drawing only roads and rivers, planners started mapping diseases. When cholera or plague appeared in crowded neighbourhoods, officials noticed that outbreaks often followed patterns: contaminated water sources, poor drainage,

and dense housing. This observation encouraged a shift from blaming individuals to examining systems. The city, in this view, was not merely a collection of people; it was an interconnected organism where pipes, wells, waste, and movement could spread health or harm. Yet the rise of public health planning was not purely scientific. It was also political. Some authorities used disease maps to justify demolishing informal settlements while ignoring the factories or elite quarters that produced pollution. In other cases, reformers pushed for safer water, vaccination drives, and sanitation laws that benefited most citizens, including the poor. The same data could support either improvement or exclusion, depending on who interpreted it and whose voices were heard. Over time, public health measures expanded

from emergency responses to long-term reforms. Clean water systems reduced water-borne diseases; waste management

lowered infection risks; and regulations improved food safety. Still, these gains were uneven. Communities with less political power often received fewer services, even when their needs were greater. The history of urban health therefore teaches a complex lesson: evidence can guide progress, but fairness depends on how evidence is used.

True

improvement requires not only technology and statistics, but also accountability, public participation, and equal access to basic services.

- A. Urban disease mapping was useless because outbreaks were random
- B. The passage explains how disease mapping shaped urban reform while also being used politically, producing uneven benefits
- C. The passage argues that only vaccination mattered in improving city health
- D. The passage focuses mainly on the design of roads and bridges

Answer: B — dual use: reform tool + political misuse, uneven gains

Trick: The passage's spine is a TWO-SIDED claim ("Yet... not purely scientific"): the main idea must contain both sides. Option B is the only one holding reform AND political misuse together.

Solution: Paras 1–2: mapping revealed system-level causes and drove reform, but was also weaponised against informal settlements. Para 3: gains uneven; fairness depends on use. B covers all three.

Trap: A and C contradict the text outright; D ignores the entire subject — the distractor set here tests whether you read the "Yet" pivot.

Q.41 Which detail best supports the claim that health planning could become a tool of exclusion?

Read the passage and answer the following questions.

In the early decades of the 20th century, a new kind of “map” began to reshape how cities were governed. Instead of drawing only roads and rivers, planners started mapping diseases. When cholera or plague appeared in crowded neighbourhoods, officials noticed that outbreaks often followed patterns: contaminated water sources, poor drainage,

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A. Officials observed that outbreaks followed patterns linked to systems

B. Some authorities used disease maps to justify demolishing informal settlements while ignoring sources of pollution elsewhere

C. Clean water systems reduced water-borne diseases

D. Regulations improved food safety

Answer: B — selective demolition while ignoring elite/industrial polluters

Trick: “Exclusion” = benefiting some while targeting others with the SAME data. Find the option with an asymmetric application (demolish the poor areas, ignore the factories).

Solution: The passage’s exact sentence — maps justified demolishing informal settlements while factories/elite quarters producing pollution were ignored — is option B.

Trap: A is neutral analysis (inclusion of evidence, not exclusion); C and D are benefits. Only B has the victim/beneficiary asymmetry the word demands.

Q.42 Which situation best applies the passage’s lesson about “evidence” and “fairness”?

Read the passage and answer the following questions.

In the early decades of the 20th century, a new kind of “map” began to reshape how cities were governed. Instead of drawing only roads and rivers, planners started mapping diseases. When cholera or plague appeared in crowded neighbourhoods, officials noticed that outbreaks often followed patterns: contaminated water sources, poor drainage,

and dense housing. This observation encouraged a shift from blaming individuals to examining systems. The city, in this view, was not merely a collection of people; it was an interconnected organism where pipes, wells, waste, and movement could spread health or harm. Yet the rise of public health planning was not purely scientific. It was also political. Some authorities used disease maps to justify demolishing informal settlements while ignoring the factories or elite quarters that produced pollution. In other cases, reformers pushed for safer water, vaccination drives, and sanitation laws that benefited most citizens, including the poor. The same data could support either improvement or exclusion, depending on who interpreted it and whose voices were heard. Over time, public health measures expanded

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A. A city bans all migration because migrants “cause” disease

B. A local leader refuses to collect data to avoid controversy

C. A city uses outbreak data to improve water lines in all wards and publicly audits spending to prevent biased allocation

D. A city uses disease maps only to increase police patrols in poor areas

Answer: C — data for universal improvement + accountability

Trick: "Applies the lesson" questions need BOTH halves of the final sentence: evidence used for progress AND fairness safeguards (accountability, participation, equal access). Option C is the only one carrying both.

Solution: Passage prescription: "accountability, public participation, and equal access." C = data + all wards + public audit — exactly that bundle.

Trap: D uses evidence too but drops fairness (patrols = the misuse pattern); a partial match loses to the option containing both halves.

Q.43 What is the organisational structure of the passage?

Read the passage and answer the following questions.

In the early decades of the 20th century, a new kind of "map" began to reshape how cities were governed. Instead of drawing only roads and rivers, planners started mapping diseases. When cholera or plague appeared in crowded neighbourhoods, officials noticed that outbreaks often followed patterns: contaminated water sources, poor drainage,

and dense housing. This observation encouraged a shift from blaming individuals to examining systems. The city, in this view, was not merely a collection of people; it was an interconnected organism where pipes, wells, waste, and movement could spread health or harm. Yet the rise of public health planning was not purely scientific. It was also political. Some authorities used disease maps to justify demolishing informal settlements while ignoring the factories or elite quarters that produced pollution. In other cases, reformers pushed for safer water, vaccination drives, and sanitation laws that benefited most citizens, including the poor. The same data could support either improvement or exclusion, depending on who interpreted it and whose voices were heard. Over time, public health measures expanded

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A. Compare and contrast of two empires' health systems

B. Chronological listing of medical discoveries

C. Cause and effect with evaluation: mapping led to reforms, but political interpretation caused unequal outcomes

D. A myth retold in sequential episodes

Answer: C — cause–effect with evaluation

Trick: Trace the connector words: "This observation encouraged..." (cause→effect), "Yet... was also political" (second effect), "Still, these gains were uneven... therefore teaches" (evaluation). Signal words classify the structure.

Solution: Mapping (cause) → reforms (effect) → political interpretation → unequal outcomes (evaluated effect with judgement). That is cause-and-effect WITH evaluation = C.

Trap: Calling it "chronological" because decades are mentioned — time markers frame the account but the LOGIC linking paragraphs is causal, not sequential listing.

Q.44 The tone of the passage is best described as:

Read the passage and answer the following questions.

In the early decades of the 20th century, a new kind of "map" began to reshape how cities were governed. Instead of drawing only roads and rivers, planners started mapping diseases. When cholera or plague appeared in crowded neighbourhoods, officials noticed that outbreaks often followed patterns: contaminated water sources, poor drainage,

and dense housing. This observation encouraged a shift from blaming individuals to examining systems. The city, in this view, was not merely a collection of people; it was an interconnected organism where pipes, wells, waste, and movement could spread health or harm. Yet the rise of public health planning was not purely scientific. It was also political. Some authorities used disease maps to justify demolishing informal settlements while ignoring the factories or elite quarters that produced pollution. In other cases, reformers pushed for safer water, vaccination drives, and sanitation laws that benefited most citizens, including the poor. The same data could support either improvement or

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A. Fearful and alarming

B. Balanced and analytical

C. Mocking and humorous

D. Highly emotional and personal

Answer: B — balanced and analytical

Trick: Tone = weight given to both sides + absence of loaded words. "In other cases," "Still," "therefore" are balancing/analytical connectives; check for any sarcasm or exclamation — none exists.

Solution: The passage credits reforms AND exposes misuse, then concludes with a reasoned prescription — measured on both sides = balanced and analytical.

Trap: Reading the criticism of authorities as "fearful/alarming" — criticism with qualifiers ("in other cases," "depending on who interpreted it") is analysis, not alarm.

Q.45 The mean of a data is 76 and its median is 100. The mode (using empirical relation) of the data is:

A. 148

B. 60

C. 174

D. 77

Answer: A — 148

Trick: Empirical formula — Mode = 3 Median – 2 Mean. Memorise as "3M – 2m" (capital = median). One-line substitution, no data needed.

Solution: Mode = 3(100) – 2(76) = 300 – 152 = 148.

Trap: Using Mode = 3 Mean – 2 Median (150 – 200 = –50, or mis-ordered as $2 \times 100 - 3 \times 76 = 228$) — the coefficient 3 belongs to the MEDIAN.

Q.46 If the first term of a geometric progression is 2 and the common ratio is 3, then what will be the fifth term of the geometric progression?

A. 243

B. 324

C. 81

D. 162

Answer: D — 162

Trick: n -th term = $a \cdot r^{(n-1)}$; "fifth" means power 4, not 5. Chain the terms by multiplication — 2, 6, 18, 54, 162 — and never touch exponents if unsure.

Solution: $T_5 = 2 \times 3^4 = 2 \times 81 = 162$.

Trap: Using 3^5 ($T_6 = 486$ or $2 \times 243 = 486$, off-list) or off-by-one chains ending at 54 (T_4); option A (243) baits pure 3^5 without the leading 2.

Q.47 Pipe A can fill a tank in 16 hours, pipe B can fill the same tank in 28 hours and pipe C can fill the same tank in 8 hours. The time taken by them to fill the same tank if they operate together is:

A. $4 \frac{12}{25}$ hours

B. $2 \frac{12}{25}$ hours

C. $8 \frac{12}{25}$ hours

D. $11 \frac{12}{25}$ hours

Answer: A — $4 \frac{12}{25}$ hours

Trick: LCM-units: capacity = LCM(16, 28, 8) = 112 units. Rates become 7, 4, 14 units/h — integers, sum 25, time = $112/25$. Convert once: $112/25 = 4 + 12/25$.

Solution: $112/16 = 7$, $112/28 = 4$, $112/8 = 14$; total 25/h. Time = $112/25 = 4 \frac{12}{25}$ h.

Trap: Adding the TIMES instead of rates ($16 + 28 + 8$ or their reciprocal mishandling), or computing $112/25 \approx 4.48$ and misreading the mixed fraction as $4 \frac{48}{100} \rightarrow 4 \frac{12}{25}$ needs 12, not 48: the digits shift if you divide $112 \div 25$ carelessly.

Q.48 Find the value of $\sqrt{(212 + \sqrt{(139 + \sqrt{900}))})}$.

- A. 25
- B. 15
- C. 20
- D. 12

Answer: B — 15

Trick: Nested radicals unravel from the INSIDE: $\sqrt{900} = 30$ (perfect square anchor), then $139 + 30 = 169 = 13^2$, then $212 + 13 = 225 = 15^2$. Each inner result is engineered to be a perfect square.

Solution: $\sqrt{900} = 30 \rightarrow 139 + 30 = 169 \rightarrow \sqrt{169} = 13 \rightarrow 212 + 13 = 225 \rightarrow \sqrt{225} = 15$.

Trap: Starting from the outside ($\sqrt{212} \approx 14.56$ first) and drowning in decimals; also stopping one layer early (returning 13 or $\sqrt{225}$ misread as 25 — option A baits $\sqrt{625}$ -style slips).

Q.49 H, O, L, D, E, and R live on six different floors of the same building. The lowermost floor in the building is numbered 1, the floor above it, number 2 and so on, till the topmost floor is numbered 6. No one lives above L. Only three people live between L and R. H lives on a floor below R. E lives on an even numbered floor. O lives on an odd numbered floor but not on floor number 5. How many people live between D and R?

- A. Zero
- B. Two
- C. Three
- D. One

Answer: B — Two

Trick: Anchor L at 6 (nobody above), then "three between L and R" puts R at 2. H below R $\rightarrow H = 1$. Even floor left for E = 4; O odd, not 5 $\rightarrow O = 3$; D takes the last floor 5.

Solution: 6-L, 5-D, 4-E, 3-O, 2-R, 1-H. Between D (5) and R (2): floors 3 and 4 = 2 people.

Trap: Placing O on 5 (explicitly barred) forcing D elsewhere, or counting "between" inclusively (3 people) instead of strictly between — the answer options are built around both slips.

Q.50 The table below gives the number of students passing an exam in a particular town.

Year Girls / Boys — 2016: 128734 / 115526; 2017: 130567 / 124313; 2018: 129209 / 122131

Find the approximate average pass percentage in all three years, if the number of students appearing for the exam in any given year is 354000.

- A. 74%
- B. 68%
- C. 71%
- D. 73%

Answer: C — 71%

Trick: Average pass % over equal-sized cohorts = TOTAL passed $\div$ TOTAL appeared — one division instead of three. Round the yearly sums to nearest hundred to keep arithmetic mental: $\approx 2,44,300 + 2,54,900 + 2,51,300 \approx 7,50,500$.

Solution: Total passed = $244260 + 254880 + 251340 = 750480$. Total appeared = $354000 \times 3 = 1062000$. % = $750480/1062000 \times 100 \approx 70.67\% \approx 71\%$.

Trap: Averaging the three yearly percentages separately and rounding each first (74.07, 71.99, 70.99 $\rightarrow$ mean 72.35 $\rightarrow$ drifts toward 73%) — pool-then-divide is both faster and safer; also mis-splitting the compressed table digits (2016128734...) wrecks every sum.

Q.51 A, B, C, D, E, F and G are sitting around a circular table facing the centre. F sits to the immediate right of G. B sits third to the left of G. A is the immediate neighbour of D and G. C sits second to the left of B. How many people sit between E and D when counted from the left of D?

- A. One
- B. Four
- C. Two
- D. Three

Answer: A — exactly one person (B) sits between D and E counting from D's left.

Trick: Build the G-block first — F on G's immediate right, B third to G's left, C second to B's left — then A must hug

BOTH G and D, forcing the unique clockwise chain G–A–D–B–E–C–F.

Solution: Seat G anywhere: F right of G, B third left, C second left of B. A needs neighbours G and D → slots G–A–D. E takes the last gap. From D's left: B, then E → 1 person between.

Trap: Counting from D's RIGHT gives 4 (option B) — the question pins the side: left of D.

Q.52 Income of Amit in 2019 was ₹22,000. He gets an increment of 20% every year. What was his income (in ₹) in 2021?

- A. 26,400
- B. 30,800
- C. 31,680
- D. 22,000

Answer: C — 20% compounded twice: $\times 1.2 \times 1.2 = \times 1.44 \rightarrow ₹31,680$.

Trick: Successive change: two +20% hits multiply as $1.2 \times 1.2 = 1.44$ (44% total, not 40%).

Solution: $22000 \times 1.44 = 31,680$. (2019→2020: 26,400; 2020→2021: 31,680.)

Trap: Simple-increment habit gives $22,000 + 2 \times 4,400 = 30,800$ (option B) — each year's increment compounds on the enhanced base.

Q.53 Yash travels 200 km at 25 km/hr, the next 210 km at 21 km/hr and the next 204 km at 51 km/hr. What is his average speed (in km/hr) for the whole journey? (Round off your answer to two decimal places)

- A. 30.01
- B. 27.22
- C. 27.91
- D. 18.68

Answer: C — 614 km in exactly 22 h → 27.91 km/h.

Trick: Average speed = total distance ÷ total TIME; pick legs whose times come out whole: $200/25$, $210/21$, $204/51 = 8, 10, 4$ h.

Solution: Total time = $8 + 10 + 4 = 22$ h; $614/22 = 27.909... \approx 27.91$.

Trap: Averaging the three speeds (≈ 32.3) or pairing a distance with the wrong speed — divide each leg separately, then sum times.

Q.54 Number 0.232323 can be written in rational form as:

- A.
- B.
- C.
- D.

(Options lost in PDF extraction.)

Answer: A — $0.232323... = 23/99$.

Trick: Pure repeating decimal: block of 2 repeating digits → denominator 99, numerator = the block (23).

Solution: $y = 0.232323...; 100y - y = 23 \rightarrow 99y = 23 \rightarrow y = 23/99$.

Trap: Writing 999 (or 990) below — count the repeating digits: a 2-digit block means two 9s.

Q.55 2 times the present age of X is 7 years less than 6 times the present age of Y. At present, P is 2 times as old as X, and Y is 8 years younger than Q. If P is 50 years older than Q, then the present age (in years) of Y is:

- A. 18
- B. 13
- C. 5
- D. 12

Answer: B — the two 2X-expressions collide: $6Y - 7 = Y + 58 \rightarrow Y = 13$.

Trick: Route everything through Y: $P = 2X$ and $P = Q + 50 = (Y + 8) + 50 \Rightarrow 2X = Y + 58$; set equal to $2X = 6Y - 7$ and subtract.

Solution: $2X = 6Y - 7$ and $2X = Y + 58 \rightarrow 5Y = 65 \rightarrow Y = 13$.

Trap: Sign slips ("7 years less" flipped to +7, or adding $58 + 7$ as 60 → option 12) — hold both 2X expressions and subtract in one step.

Q.56 Based on the English alphabetical order, three of the following four letter-cluster pairs are alike in a certain way and thus form a group. Which letter-cluster pair DOES NOT belong to that group?

(Note: The odd one out is not based on the number of consonants/vowels or their position in the letter-cluster.)

- A. KN-GI
- B. SV-OQ
- C. OR-KM
- D. VY-RS

Answer: D — VY→RS steps -4, -6; the other three step -4, -5.

Trick: Convert each pair to letter-shifts; A/B/C all give (-4, -5), so the pair breaking the SECOND shift is the odd one.

Solution: KN→GI: -4, -5 · SV→OQ: -4, -5 · OR→KM: -4, -5 · VY→RS: -4, -6 → D.

Trap: Verifying only the first letter (-4 holds in all four) — the second letter's step is the discriminator.

Q.57 A sample of milk from a vessel contains 4% water. What quantity of pure milk should be added to 8 L of milk in the vessel to reduce the water content to 2%?

- A. 8 L
- B. 7 L
- C. 7.5 L
- D. 6.5 L

Answer: A — water stays 0.32 L; 2% of the new total needs 16 L → add 8 L.

Trick: Adding pure milk never changes the WATER mass — final volume = water ÷ 2% = 0.32/0.02 = 16 L.

Solution: Water = 4% of 8 = 0.32 L. Required total = 0.32 × 50 = 16 L → add 16 – 8 = 8 L. (Alligation check: 96% and 100% milk mixing to 98% → 1:1.)

Trap: Subtracting percentages (4 – 2 = 2% of 8 L) or "halving the water" — water is constant; the VOLUME doubles.

Q.58 The average weight of Vipul, Mohan, and Mohammad is 46 kg. If the average weight of Vipul and Mohan is 36 kg and that of Mohan and Mohammad is 43 kg, then the weight of Mohan (in kg) is:

- A. 30
- B. 20
- C. 35
- D. 40

Answer: B — Mohan = (V+M) + (M+Mo) – (V+M+Mo) = 72 + 86 – 138 = 20 kg.

Trick: The name repeated in both pair-sums is isolated by: pair-sum + pair-sum – trio-sum.

Solution: V+M = 72, M+Mo = 86 → V+2M+Mo = 158; minus 138 → M = 20.

Trap: Reversing the subtraction (138 – 158) or averaging 36 and 43 — add the two pairs first, then subtract the trio.

Q.59 The following are the weights (in kg) of 25 students:

58, 55, 53, 50, 53, 51, 52, 54, 53, 52, 54, 53, 58, 53, 59, 55, 53, 52, 51, 54, 53, 59, 55, 53, 52

What is the range of the given data?

- A. 6
- B. 7
- C. 9
- D. 8

Answer: C — max 59 – min 50 = 9.

Trick: Hunt only the extremes, ignore repeats: 59 appears thrice, 50 exactly once.

Solution: Highest = 59, lowest = 50 → range = 9.

Trap: Mis-scanning the lone 50 as 51 (or missing a 59) shifts the range to 8 or 7 — lock the extremes first.

Q.60 The tenth term of the sequence 2, 5, 8, 11,will be:

- A. 28
- B. 29
- C. 32
- D. 27

Answer: B — $a + (n-1)d = 2 + 9 \times 3 = 29$.

Trick: n-th term = first + (n-1) gaps; the 10th term sits 9 jumps of 3 from 2.

Solution: $d = 3$; $T_{10} = 2 + 27 = 29$.

Trap: Using $n \times d$ ($2 + 30 = 32$, option C) — the first term already occupies $n = 1$.

Q.61 The percentage profit earned by selling a table lamp for ₹863 is equal to the percentage loss incurred by selling the same table lamp for ₹433. At what price (in ₹) should the table lamp be sold to make 25% profit?

- A. 812

- B. 810
- C. 811
- D. 813

Answer: B — $CP = (863 + 433)/2 = 648$; for 25% profit, $SP = 648 \times 5/4 = 810$.

Trick: Equal profit% and loss% $\Rightarrow$ CP is the plain AVERAGE of the two selling prices; then multiply by 5/4.

Solution: $(863 + 433)/2 = 648$; $648 \times 1.25 = 810$.

Trap: Options 811/812/813 hug the answer for $\times 1.25$ slips — do it as $648 + 648/4 = 648 + 162 = 810$ exactly.

Q.62 What is the median of the following distribution?

Class

140-190 190-240 240-290 290-340 340-390

Frequency

23

13

24

25

35

A. 301

B. 278

C. 290

D. 304

Answer: C — $N/2 = 60$ falls at the close of 240–290 $\rightarrow$ median = 290.

Trick: Run cumulative frequencies 23, 36, 60, 85, 120 — the 60th value is the LAST of the third class ($60 - 36 = 24 =$ its frequency), so the median lands exactly on the upper limit 290.

Solution: $N = 120$, $N/2 = 60$; median class 240–290 ($36 < 60 \leq 60$). Median = $240 + 50 \times (60 - 36)/24 = 290$.

Trap: Pushing into 290–340 because "60 crosses 36" — the $\leq$ side keeps the 60th value inside 240–290.

Q.63 Find the next number in the given series:

9, 11, 15, 17, 21, 23, ?

A. 29

B. 28

C. 30

D. 27

Answer: D — alternate +2, +4 $\rightarrow 23 + 4 = 27$.

Trick: Check the gap cycle first: +2, +4, +2, +4... — after a +2 step the next step is +4.

Solution: $9(+2)11(+4)15(+2)17(+4)21(+2)23(+4) \rightarrow 27$.

Trap: Averaging the jumps (adding 6 $\rightarrow$ 29, option A) — the pattern alternates, it does not settle at 6.

Q.64 This question is based on the following words.

GOT HER WAR FIT

In each of the words, each vowel is changed to the letter immediately following it in the English alphabetical order and each consonant is changed to the letter immediately preceding it in the English alphabet. In how many letter-clusters thus formed will no vowel appear?

A. Three

B. One

C. Four

D. Two

Answer: A — FPS, GFQ, VBQ are vowel-free; EJS is not $\rightarrow$ three words.

Trick: Consonants step BACK, vowels step FORWARD; then re-scan the NEW strings — a shifted consonant can itself become a vowel (F $\rightarrow$ E).

Solution: GOT $\rightarrow$ FPS, HER $\rightarrow$ GFQ, WAR $\rightarrow$ VBQ (clean); FIT $\rightarrow$ EJS (E surfaced from F). Count = 3.

Trap: Reversing the direction rule yields only 1 (option B); and forgetting F $\rightarrow$ E creates a vowel miscounts FIT.

Q.65 Find the perimeter of a rectangle whose length and breadth are 5 cm and 8 cm.

A. 20 cm

B. 26 cm

C. 39 cm

D. 13 cm

Answer: B — $2(5 + 8) = 26$ cm.

Trick: Perimeter = $2(l + b)$; compute the bracket first: 13×2 .

Solution: $2 \times 13 = 26$ cm.

Trap: Stopping at $l + b = 13$ (option D) — perimeter doubles the half-round.

Q.66 Which of the following letter-clusters should replace # and % so that the pattern and relationship followed between the letter-cluster pair on the left side of :: is the same as that on the right side of :: ?

: RWS :: INJ : %

A. # = NGV, % = FLP

B. # = PIV, % = MJU

C. # = UZV, % = FKG

D. # = CDR, % = KJI

Answer: C — UZV→RWS and INJ→FKG both shift every letter by -3.

Trick: For cluster analogies, extract ONE uniform shift vector from the left pair and replay it on the right pair — only a consistent (-3, -3, -3) survives both sides.

Solution: U-3=R, Z-3=W, V-3=S ✓; the same -3 on I, N, J gives F, K, G ✓ → # = UZV, % = FKG.

Trap: Options matching just one letter's move (J-3 = G appears in several) bait a partial match — every letter must shift identically.

Q.67 Observe the table below and answer the following question:

YearPopulationConsumption of electricity (GW)

2015

20

25

2016

30

40

2017

40

60

2018

50

80

2019

60

100

1 GW = 100 Million Watt

Population in Million

In which year did electricity consumption grow by 50%?

A. 2017

B. 2018

C. 2016

D. 2019

Answer: A — 40→60 GW during 2017 is exactly +50%.

Trick: % growth = (new - old)/old; the consumption column (25, 40, 60, 80, 100) gives 60%, 50%, 33⅓%, 25% — only one is 50%.

Solution: 2016→2017: $(60 - 40)/40 = 50\%$, recorded in 2017.

Trap: Reporting the base year (2016) or dividing by the NEW value — the denominator is always the previous year's figure.

Q.68 Three numbers are in the ratio 2: 3: 4. If their LCM is 480, then their HCF is:

A. 20

B. 40

C. 80

D. 2

Answer: B — numbers $2x, 3x, 4x$ have $\text{LCM} = 12x = 480 \rightarrow x = 40 = \text{HCF}$.

Trick: For numbers in ratio $k_1:k_2:k_3$, $\text{HCF} = x$ and $\text{LCM} = \text{LCM}(k_1, k_2, k_3) \times x$; here $\text{LCM}(2, 3, 4) = 12$.

Solution: $12x = 480 \rightarrow x = 40 \rightarrow \text{HCF} = 40$.

Trap: Taking $\text{LCM}(2, 3, 4) = 24 \rightarrow$ the bait 20 (option A); 2, 3, 4 multiply through to only 12.

Q.69 For what prime number p in the number $3785p$ so that the number is exactly divisible by 3?

A. 7

B. 1

C. 3

D. 5

Answer: A — digit sum $23 + p$ must be a multiple of 3 AND p prime $\rightarrow p = 7$ ($23 + 7 = 30$).

Trick: Solve $23 + p \equiv 0 \pmod{3} \rightarrow p \in \{1, 4, 7\}$; strike out the non-primes $\rightarrow 7$.

Solution: $3+7+8+5 = 23$; $23 + 7 = 30$ ✓ divisible by 3, and 7 is prime.

Trap: Option B ($p = 1$) also makes the digit sum 24 — but 1 is NOT prime; both conditions must hold.

Q.70 Refer to the following number and symbol series and answer the question that follows.

Counting to be done from left to right only.

(Left) 1 5 & 3 * \$ 4 @ % 2 7 # 9 ^ # >= 6 8 (Right)

How many such symbols are there, each of which is immediately preceded by a number and also immediately followed by another number?

A. One

B. Two

C. None

D. Three

Answer: B — only & (between 5 and 3) and # (between 7 and 9) qualify.

Trick: Sweep once for the digit–symbol–digit frame; any symbol touching ANOTHER symbol is automatically disqualified.

Solution: & ✓ (5_3); # ✓ (7_9); , \$, @, %, ^, #, >= all touch a symbol $\rightarrow \text{count} = 2$.

Trap:* Counting \$ or % — they sit beside other symbols, so "preceded AND followed by numbers" kills them.

Q.71 Amita travels a certain distance at a speed of 84 km/hr and double the earlier distance at 28 km/hr. She then returns to the starting point through the same route. If her average speed for the entire journey is 24 km/hr, then what is her speed (in km/hr) for the return journey?

A. 17.2

B. 16

C. 18

D. 21.5

Answer: C — total $6x$ in $x/4$ hours; return leg $3x$ in $x/6 \rightarrow 18$ km/h.

Trick: Keep distances in units of x : onward time $= x/84 + 2x/28 = x/12$; total time $= 6x/24 = x/4 \rightarrow$ return time $= x/4 - x/12 = x/6$ for $3x \rightarrow v = 18$.

Solution: $1/84 + 2/28 = 1/12$ (per x). Total time $x/4 \rightarrow$ return $3x$ in $x/6 \rightarrow 18$ km/h.

Trap: Forgetting the return leg doubles the distance ($6x$, not $3x$) — the 16/21.5 baits come from such half-set equations.

Q.72 Following are the ages (in years) of 6 people in a group: 25, 30, 35, 40, 45 and 50. What is the standard deviation of their ages (rounded to two decimal places)?

A. 9.26

B. 7.38

C. 8.54

D. 6.57

Answer: C — mean 37.5, $\Sigma d^2 = 437.5$, $\sigma = \sqrt{(437.5/6)} \approx 8.54$.

Trick: Deviations are symmetric $\pm 2.5, \pm 7.5, \pm 12.5 \rightarrow$ squares 6.25, 56.25, 156.25, each twice $\rightarrow 437.5$; divide by $n = 6$, then root.

Solution: Mean $= 225/6 = 37.5$; variance $= 437.5/6 = 72.92$; $\sigma = 8.54$.

Trap: Dividing by $n-1$ (sample formula $\rightarrow 9.35$) — the question treats the six ages as the complete data (population SD).

Q.73 In a class consisting of 67 students, Teena ranked 12th from the top and her friend Rishabh ranked 26th from the bottom. How many students ranked between Teena and Rishabh?

- A. 33
- B. 27
- C. 28
- D. 29

Answer: D — Rishabh is 42nd from top; between = $42 - 12 - 1 = 29$.

Trick: Convert bottom-rank first: from-top = $N - \text{from-bottom} + 1$; then between = gap - 1.

Solution: $67 - 26 + 1 = 42$; $42 - 12 - 1 = 29$.

Trap: Skipping the +1 (using 41) lands on 28 (option C) — the classic rank-conversion slip.

Q.74 A sum of money is to be distributed among A, B, C, and D in the ratio of 14 : 7 : 11 : 10. If C gets ₹196 more than B, how much did A receive?

- A. ₹684
- B. ₹686
- C. ₹688
- D. ₹685

Answer: B — $C - B = 4$ units = ₹196 → unit ₹49; $A = 14 \times 49 = ₹686$.

Trick: For "who gets how much more" questions: one unit = given difference ÷ ratio difference; then multiply by the target's ratio.

Solution: $11 - 7 = 4$; $196/4 = 49$; $14 \times 49 = 686$.

Trap: Rushed division $196/4$ — options 684/688 flank the true 686; keep 49 exact, then $\times 14$.

Q.75 M is inversely proportional to N. If M is 18, then N is 10. If N = 9, then what is the value of M?

- A. 22
- B. 21
- C. 23
- D. 20

Answer: D — $M \times N = 180$ is constant → $M = 180/9 = 20$.

Trick: "Inversely" → multiply the known pair once: $18 \times 10 = 180$; every new case is $180 \div N$.

Solution: $M \times 9 = 180 \rightarrow M = 20$.

Trap: Treating it as direct variation ($M = 18 \times 9/10 = 16.2$) — fix the PRODUCT, then divide.

Q.76 The selling price of 34 books is equal to the cost price of 17 books. Find the loss or gain percentage.

- A. 50% loss
- B.
- C.
- D.

(Options B–D lost in PDF extraction.)

Answer: A — SP of 34 = CP of 17 → each book sells at half its CP → 50% loss.

Trick: Scale to ONE book: $SP_1 = 17C/34 = C/2$ — selling at half of cost is a straight 50% loss.

Solution: Loss per book = $C - C/2 = C/2 \rightarrow \text{loss}\% = 50$.

Trap: Flipping the equality (CP of 34 = SP of 17) turns it into a 100%-gain reading — check which side carries the bigger count.

Q.77 The 7-digit number 51A808B is divisible by 24. What is the minimum value of (A + B)?

- A. 5
- B. 3
- C. 2
- D. 6

Answer: C — $B \in \{0, 8\}$ from the 8-test; with $B = 0$, digit sum $22 + A$ forces $A \equiv 2 \pmod{3} \rightarrow A = 2 \rightarrow A + B = 2$.

Trick: Split $24 = 3 \times 8$; let the 8-test pin B first, then the mod-3 residue pins the least A.

Solution: Last three digits 08B divisible by 8 → $B = 0$ or 8; take $B = 0$. Sum = $22 + A + B \equiv 0 \pmod{3} \rightarrow A = 2$.

(Check: $5128080 \div 24 = 213670$ ✓)

Trap: Picking $B = 8$ "to be safe" inflates $A + B$, and skipping the mod-3 residue ($A = 0$ or 1 both fail) — minimise B first, then A.

Q.78 The sum of 4 consecutive odd numbers is 160. Find the smallest number.

- A. 37
- B. 27
- C. 35
- D. 25

Answer: A — four consecutive odds average $160/4 = 40 \rightarrow 37, 39, 41, 43$.

Trick: The mean of an even count of consecutive-odd numbers sits between the middle pair $\rightarrow$ smallest = mean $- 3$.

Solution: $40 - 3 = 37$ (check: $37 + 39 + 41 + 43 = 160$).

Trap: Subtracting a wrong offset from 40 (e.g., $40 - 5 = 35$, option C) — four odd neighbours span only 3 gaps of 2, so step back 3.

Q.79 If $+$ means $-$, $-$ means $\times$, $\times$ means $\div$ and $\div$ means $+$, then what will come in place of the question mark (?) in the following equation?

$$12 - 2 + 49 \times 7 \div 5 = ?$$

- A. 22
- B. 27
- C. 12
- D. 17

Answer: A — mapped line: $12 \times 2 - 49 \div 7 + 5 = 24 - 7 + 5 = 22$.

Trick: Rewrite the ENTIRE expression with the new symbols first, then apply BODMAS — never substitute mid-calculation.

Solution: $12 \times 2 = 24$; $49 \div 7 = 7$; $24 - 7 + 5 = 22$.

Trap: Swapping only one sign pair or evaluating in written order — complete the substitution, then let BODMAS sequence it.

Q.80 Hemant and Irfan can complete a certain piece of work in 6 and 10 days, respectively. They started to work together, and after 2 days, Irfan left. In how many days will Hemant complete the remaining work?

- A. $2 \frac{2}{5}$ days
- B. $3 \frac{1}{5}$ days
- C. $2 \frac{1}{5}$ days
- D. $2 \frac{4}{5}$ days

(Option values reconstructed from the original solution's $14/5$ -day result; PDF extraction garbled them.)

Answer: D — together 2 days finish 16 of 30 units; Hemant alone does the last 14 in $14/5 = 2 \frac{4}{5}$ days.

Trick: LCM 30 units: H = 5/day, I = 3/day $\rightarrow$ 2 joint days = 16 units; remaining $14 \div 5$ (Hemant's rate) = $2 \frac{4}{5}$.

Solution: $30 - 16 = 14$ units left; $14/5 = 2.8$ days = $2 \frac{4}{5}$ days.

Trap: Subtracting 2 joint days from 6 ($\rightarrow$ 4 days, option-bait style) — Irfan's 3 units/day also consumed work in those 2 days.

Q.81 The arithmetic mean of the observations 28, 31, 40, 63, 57, 37, 34, 70 and 99 is:

- A. 55
- B. 50
- C. 41
- D. 51

Answer: D — sum 459 over 9 values $\rightarrow 51$.

Trick: Cluster the addition: $(28+31+40) = 99$, $(63+57) = 120$, $(37+34+70) = 141$, $+99 \rightarrow 459$.

Solution: $459/9 = 51$.

Trap: A mis-added sum or dividing by 8 — there are nine observations and the sum is exactly 459.

Q.82 What will come in place of the question mark (?) in the following equation, if $+$ and $-$ are interchanged and $\times$ and $\div$ are interchanged?

$$28 + 125 \times 5 - 12 \div 3 = ?$$

- A. 28
- B. 46
- C. 39
- D. 21

Answer: C — after both swaps: $28 - 125 \div 5 + 12 \times 3 = 28 - 25 + 36 = 39$.

Trick: Swap BOTH pairs in one rewrite, then BODMAS with strict left-to-right for the surviving $+$ and $-$.

Solution: $125 \div 5 = 25$; $12 \times 3 = 36$; $28 - 25 + 36 = 39$.

Trap: Interchanging only one pair, or folding $28 - 25 + 36$ as $28 - (25 + 36) = -33$ — resolve $\div$ and $\times$ first, then move left to right.

Q.83 The simplified value of $72 - (-98) \times (-31 - 58 - 19) \div [9 \times \{7 + (-2) \times (-6)\}]$ is:

- A.
- B.
- C.
- D.

(Options lost in PDF extraction.)

Answer: A — $72 - 10584/171 = 10 \frac{2}{19} = 192/19$.

Trick: Clear the signs first: $(-31 - 58 - 19) = -108$, $(-98) \times (-108) = +10584$, inner $\{7 + 12\} = 19 \rightarrow$ one clean division.

Solution: $72 - 10584 \div 171 = 72 - 61 \frac{17}{19} = 10 \frac{2}{19} = 192/19 (\approx 10.11)$.

Trap: One sign slip — $(-2) \times (-6)$ as -12 , or dropping the minus before (-98) — shifts the numerator onto the 182/187/188-style distractors.

Q.84 A alone can complete a piece of work in 15 days, while B alone can do it in 30 days, if the both of them work on alternate days, when will the work get completed?

- A. Cannot be determined
- B. 24 days
- C. 20 days
- D. 18 days

Answer: C — each 2-day cycle completes $1/10$; exactly 10 cycles $\rightarrow$ 20 days.

Trick: Take LCM 30 units: A delivers 2/day, B 1/day $\rightarrow$ 3 units per 2-day cycle; $30 \div 3 = 10$ cycles.

Solution: $10 \times 2 = 20$ days; the last day (B's) lands the total exactly on 30 units — no fractional day.

Trap: Averaging 15 and 30, or fearing a partial final day — the cycle divides the job perfectly at 20.

Q.85 In a certain code language,

'P + Q' means 'P is the son of Q',

'P - Q' means 'P is the wife of Q',

'P $\times$ Q' means 'P is the husband of Q' and

'P $\div$ Q' means 'P is the brother of Q',

How is A related to E if 'A + B - C + D $\times$ E'?

- A. Son's son
- B. Daughter's son
- C. Brother
- D. Son

Answer: A — A is B's son; B is C's wife; C is D's son; D is E's husband $\Rightarrow$ A is E's son's son.

Trick: Read the chain left to right as generation links; spouse symbols ($-$, $\times$) weld the couple together so their children are shared.

Solution: A(+B): A son of B $\rightarrow$ B(-C): B wife of C $\rightarrow$ A is C's son $\rightarrow$ C(+D): C son of D $\rightarrow$ D($\times$ E): D husband of E $\rightarrow$ C is E's son. So A = son of E's son.

Trap: Treating B-C as "B's mother" or splitting the couple (keeping A outside C's line) — the standard convention merges spouses' children.

Q.86 20 men and 15 boys can do a piece of work in 10 days. 25 men and 10 boys can do it in 9 days. find the ratio of the daily work done by a man to that of a boy.

- A. 14 : 5
- B. 5 : 12
- C. 12 : 5
- D. 5 : 14

Answer: C — $(20M + 15B) \times 10 = (25M + 10B) \times 9 \rightarrow 60B = 25M \rightarrow M : B = 12 : 5$.

Trick: Equate the two total-work expressions and let the coefficients collapse — no LCM or units needed.

Solution: $200M + 150B = 225M + 90B \rightarrow 60B = 25M \rightarrow M/B = 12/5$.

Trap: Inverting at the finish (5 : 12, option B) — the question asks man : boy.

Q.87 According to the following graph, find the average annual percentage growth of mascara through the years

2001, 2002, 2003 and 2004.

- A. 75%
- B. 50%
- C. 100%
- D. 46%

[Figure note: the graph did not survive PDF extraction; the original solution used 2001 = 2 and 2004 = 6 (in hundreds).]

Answer: D — CAGR from 2 (2001) to 6 (2004) over 3 years $\approx 44.2\%$ p.a.; nearest option 46%.

Trick: Average annual % growth across n years = $(\text{End}/\text{Start})^{(1/n)} - 1$; here $(6/2)^{(1/3)} - 1 = 3^{(1/3)} - 1$.

Solution: $3^{(1/3)} \approx 1.442 \rightarrow \approx 44.2\%$ per year (the key rounds the root to 1.46); the only option in range is 46% (D).

Trap: Reading "average growth" as total $200\% \div 3 \approx 67\%$ or compounding year-by-year guesses — use the root form and pick the nearest option.

Q.88 A cylindrical rod has an outer curved surface area of 1800 cm^2 . If the length of the rod is 52 cm, then the outer radius (in cm) of the rod, correct to two places of decimal, is:

Take $\pi = 22/7$.

- A. 6.32
- B. 5.51
- C. 5.37
- D. 6.69

Answer: B — $r = 1800 \times 7 / (2 \times 22 \times 52) = 12600/2288 \approx 5.51 \text{ cm}$.

Trick: $\text{CSA} = 2\pi rh \rightarrow r = \text{CSA}/(2\pi h)$; with $\pi = 22/7$ everything collapses to one fraction: $12600/2288$.

Solution: $2 \times (22/7) \times r \times 52 = 1800 \rightarrow r = 5.507 \approx 5.51$.

Trap: Dropping the factor 2 (using πrh) or reading 52 as the diameter — the formula wants radius and the full $2\pi rh$.

Q.89 The length of each of the two equal sides of an isosceles triangle is 41 cm and the length of its base is 18 cm.

The area (in cm^2) of the triangle is:

- A. 351
- B. 355
- C. 360
- D. 365

Answer: C — the 9-40-41 right triangle: height 40, area = $\frac{1}{2} \times 18 \times 40 = 360$.

Trick: The altitude bisects the base: $h = \sqrt{(41^2 - 9^2)} = \sqrt{1600} = 40$ — spot the Pythagorean triple and skip the arithmetic.

Solution: Half base = 9; $h = 40$; area = $9 \times 40 = 360 \text{ cm}^2$.

Trap: Putting the FULL base 18 under the root ($\sqrt{(1681 - 324)}$) — the altitude's foot bisects an isosceles base.

Q.90 What is the main idea of the passage?

Read the passage and answer the following questions.

Many myths describe a human who tries to steal a divine secret—fire, immortality, or forbidden knowledge—and pays a price. Such stories are often treated as warnings against arrogance. But they also function like social instruction manuals. By turning curiosity into a dramatic plot, myths teach communities how to manage new powers without destroying trust. When a society faces a disruptive skill—metalworking, navigation, writing, or modern technology—it must decide who may use it, under what rules, and with what responsibility. Myths compress these debates into memorable scenes.

Consider the recurring pattern: a gift is taken from a god; the gift improves life; and the punishment defines boundaries. The gift explains progress, while the punishment explains restraint. Importantly, the restraint is not always purely moral. It can be political: rulers may present themselves as guardians who control dangerous knowledge. At the same time, myths can protect ordinary people by demanding that powerful individuals face limits. A story that punishes the overreaching hero may be less about rejecting innovation and more about placing innovation inside a framework of accountability.

This is why myth remains relevant even in technological cultures. When people debate artificial intelligence, genetic editing, or surveillance, they often repeat mythic questions: Who benefits? Who bears the risk? Who decides what is "too much"? The language may be modern, but the structure is old. Myths, then, are not frozen fantasies. They are tools for thinking—systems for balancing desire for advancement with fear of harm, and for turning complex social problems into narratives that communities can remember and argue about.

A. Myths are childish stories that have no purpose today

- B. The passage argues that myths about stealing divine secrets function as social frameworks that balance innovation with accountability, and they still shape modern debates
- C. The passage claims that all myths were created only to entertain rulers
- D. The passage proves that technology always leads to punishment

Answer: B — myths act as social frameworks balancing innovation with accountability, and still shape modern debates.

Trick: Main idea = the full arc (function + modern link). Para 2 hands you "accountability", para 3 hands you "AI/genetic editing/surveillance" — the option joining both is B.

Solution: B restates the framework role AND the modern relevance; A, C, D each contradict an explicit line (myths remain relevant; rulers are one user, not the purpose; punishment sets boundaries rather than dooming technology).

Trap: Absolute words — "childish", "no purpose", "only", "always" — none is supported anywhere in the passage; treat extremes as auto-rejects.

Q.91 Which detail best supports the passage's claim that myths "compress debates" into stories?

Read the passage and answer the following questions.

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A. Myths always include magical animals

B. Myths turn questions about who may use a disruptive skill and under what rules into memorable scenes of gifts and punishments

C. Myths are written only in poetic language

D. Myths avoid politics completely

Answer: B — gift → benefit → punishment scenes ARE the compressed debate about who may use a skill and under what rules.

Trick: "Best supports" demands the detail that mirrors the quoted mechanism, not merely a true statement — match scene-structure to "compress into memorable scenes".

Solution: B converts the societal debate (who may use, under what rules, with what responsibility) into the gift/punishment plot — exactly the compression claim.

Trap: Picking politically flavoured but tangential details (rulers as guardians) — they support a side point, not the compression mechanism.

Q.92 Which situation best reflects the passage's interpretation of "punishment" in these myths?

Read the passage and answer the following questions.

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- A. A story punishes a hero to prove curiosity is always evil
- B. A society uses a cautionary tale to set rules so that powerful tools are used under accountability rather than without limits
- C. A community bans all learning to avoid risk
- D. A ruler encourages unrestricted access to surveillance tools because myths celebrate power

Answer: B — punishment = boundary-setting so powerful tools stay under accountability.

Trick: Carry the passage's reframe ("less about rejecting innovation, more about accountability") into the options and pick the one with rules + accountability.

Solution: B: cautionary tale → rules → accountable use of powerful tools. A, C, D distort the idea (curiosity evil, blanket bans, unrestricted power).

Trap: Option A's "curiosity is always evil" over-moralises — the passage explicitly says the restraint is not always purely moral.

Q.93 What is the organisational structure of the passage?

Read the passage and answer the following questions.

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- A. Sequential steps for writing a myth
- B. Compare and contrast between two mythological characters only
- C. Description with cause-effect reasoning: myths shape social rules around innovation and reappear in modern tech debates
- D. Chronological account of myth creation century by century

Answer: C — description plus cause-effect: myths shape rules around innovation and resurface in modern debates.

Trick: Name the structure from its connectives: "Consider the recurring pattern... This is why myth remains relevant..." = description feeding cause-effect evaluation.

Solution: C captures both halves (social-rule shaping + modern reappearance); the others invent steps, timelines, or two-character contrasts that never appear.

Trap: "Chronological/sequential" bait — no timeline or how-to sequence exists anywhere in the passage.

Q.94 The style of the passage is best described as:

Read the passage and answer the following questions.

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A. Argumentative and interpretive, using patterns to connect mythology with modern culture

B. Purely scientific with formulas and data tables

C. Comedic and satirical

D. Personal diary writing

Answer: A — argumentative-interpretive: recurring myth patterns mapped onto modern culture.

Trick: Judge style by HOW the passage argues, not what it says: pattern analysis + modern application = interpretive argument.

Solution: A fits (patterns → mythology-to-modernity bridge); there are no formulas/tables (B), no satire (C), no diary voice (D).

Trap: Confusing "analytical" with "purely scientific" — the passage cites no data or formulas; its method is interpretive.

Q.95 P and Q started a business with equal capital. P withdrew from the business at the end of 4 months. If, at the end of the year, the business made a profit of 6,400, then the share of P is:

A. Rs. 1,600

B. Rs. 3,200

C. Rs. 1,800

D. Rs. 1,750

Answer: A — equal capital $\Rightarrow$ profit ratio = time ratio $4 : 12 \rightarrow P = 6400 \times 4/16 = 1600$.

Trick: With identical investments, months alone decide shares: $P:Q = 4:12$; total parts 16, P holds 4.

Solution: $P = 6400 \times 4/16 = 1600$.

Trap: Counting P's months as 6 (half the year) $\rightarrow$ the 3,200 bait (option B); he left at the END of 4 months.

Q.96 If the length of a cuboid is 33 cm, breadth is 4 cm and volume is 7524 cm^3 , then find the height (in cm) of the cuboid.

A. 60

B. 57

C. 54

D. 67

Answer: B — $h = 7524 \div (33 \times 4) = 7524/132 = 57$.

Trick: Volume $\div (l \times b)$; use the divisibility check — $132 \times 57 = 7524$ exactly, no long division.

Solution: $33 \times 4 = 132$; $7524/132 = 57 \text{ cm}$.

Trap: Adding $l + b$ (37) or fumbling 132 — $7524 \div 132$ is exactly 57.

Q.97 Seven people, K, L, M, N, O, P, and Q are sitting in a straight line facing north. Only three people sit between L and O. Only two people sit to the right of O. K sits second to the right of P. K is not an immediate neighbour of Q. O sits to the immediate left of M. Who sits at the extreme left end of the line?

A. N

B. P

C. M

D. L

Answer: D — the unique line is L P N K O M Q $\rightarrow$ extreme left = L.

Trick: Anchor on O: "only two to the right" fixes O at slot 5 of 7; the 3-between rule then drops L at slot 1 and M at 6

— half the seat map done in two steps.

Solution: O5 → L1, M6. $K = P + 2$: P2→K4 works (P3→K5 clashes with O). Q cannot neighbour K4 → Q7, leaving N3. Leftmost = L.

Trap: Reversing "O sits to the immediate left of M" (placing O right of M) leaves Q no legal seat — dead ends push guessers toward N/P/M.

Q.98 Read the given data and answer the question that follows.

2, 5, 15, 25, 20, 12, 8, 7, 6, 16, 21, 17, 30, 32, 23, 40, 51, 15, 2, 9, 57, 19, 25

If the given data is grouped in the classes 0-5, 5-10, 10-15, and so on, then what will be the frequency of the class 20-25?

- A. 4
- B. 3
- C. 5
- D. 2

Answer: B — values with $20 \leq x < 25$ are 20, 21, 23 → frequency 3.

Trick: Classes are [a, b): the LOWER limit joins the class, the UPPER limit migrates to the next one.

Solution: In-range: 20, 21, 23 (the 25s belong to 25–30) → 3.

Trap: Counting 25 into 20–25 gives the bait 4 (option A) — upper limits always move right in exclusive classes.

Q.99 A box contains 6 white, 2 black and 3 red balls. If a ball is drawn at random, what is the probability that it will not be white?

- A.
- B.
- C.
- D.

(Options lost in PDF extraction.)

Answer: C — non-white = $2 + 3 = 5$ out of 11 → $5/11$.

Trick: "Not white" = add the non-white counts directly ($2 + 3$); or $1 - 6/11$ — never touch the white count as numerator.

Solution: $P(\text{not white}) = 5/11$.

Trap: Answering $P(\text{white}) = 6/11$ — the negation flips the NUMERATOR pool, not the denominator.

Q.100 Refer to the following number and symbol series and answer the question that follows.

Counting to be done from left to right only.

(NOTE: All numbers are single digit numbers only.)

(Left) 7 @ £ 1 @ 7 # * 8 7 8 6 5 3 8 4 € # £ 9 1 £ (Right)

How many such symbols are there each of which is immediately preceded by a number and also immediately followed by a number?

- A. 1
- B. 3
- C. 2
- D. 0

Answer: A — only the second @ (between 1 and 7) is number–symbol–number.

Trick: One sweep with the digit–symbol–digit frame; every £/#/ here touches another symbol, so they die instantly.

Solution: @ after 1 and before 7 ✓; the first @ (7_£), #, , €, #, £ all fail the both-sides test → count = 1.

Trap: Auto-accepting the first @ — its right neighbour is £, a symbol, not a number.